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relation between determinant of A and determinant of adj A and eterminant of inverse of A where A is a n x n matrix

For an $n \times n$ matrix A, there are important relationships between:

- $\det(A)$
- $\det(\text{adj}\, A)$
- $\det(A^{-1})$

(assuming A is **invertible**, so $\det(A) \neq 0$).

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1. Determinant of the inverse

If A is invertible,

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

⭐ Reason:

From the property

$$\det(AB) = \det(A)\det(B)$$

Since

$$AA^{-1} = I$$

Taking determinant:

$$\det(A)\det(A^{-1}) = \det(I) = 1$$

So

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

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2. Determinant of the adjoint

For an $n \times n$ matrix:

$$\det(\operatorname{adj} A) = (\det A)^{n-1}$$

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3. Relation between adjoint and inverse

Another important identity:

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

This connects all three quantities.

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4. Combined relationship

Using the inverse formula:

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

Taking determinant of both sides:

$$\det(A^{-1}) = \det\left(\frac{1}{\det(A)} \operatorname{adj}(A)\right)$$

Since multiplying a matrix by scalar c multiplies determinant by c^n:

$$\det(A^{-1}) = \left(\frac{1}{\det(A)}\right)^n \det(\operatorname{adj} A)$$

Substitute $\det(\operatorname{adj} A) = (\det A)^{n-1}$:

$$\det(A^{-1}) = \frac{(\det A)^{n-1}}{(\det A)^n}$$

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

✔ confirms the previous result.

Final Key Formulas 📌

$$\boxed{\det(A^{-1}) = \frac{1}{\det(A)}}$$

$$\boxed{\det(\text{adj } A) = (\det A)^{n-1}}$$

$$\boxed{A^{-1} = \frac{1}{\det(A)} \text{ adj}(A)}$$

If you want, I can also show a **quick exam shortcut trick professors expect in Linear Algebra papers** (takes 10 seconds to write in exams) 📘.